Whitepaper for a trusty cam - a block-chain enabled AI smart camera supporting privacy connected to an open marketplace interface

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***Abstract*— Camera technology improves continuously with higher resolution, higher frame rates, miniaturization, 3D capability, 360 degree mechanics and the accuracy in image detection (driven by AI) is close to or beating human level. In contrary, privacy concerns and privacy requirements increase. This paper describes how block-chain and minor hardware adjustments for privacy can strengthen trust, ensure transparency and extend business models for smart cameras if they get connected to a marketplace for computer vision.**

Keywords: block-chain video camera, smart camera, decentralized video camera, mobile video camera, closed circuit television, CCTV, anonymous video camera, privacy, AI marketplace, computer vision, business models, surveillance systems, distributed event-based systems

I. History

The first video camera was invented by Louis Le Prince and published 1888. His *moving images* are capturing the popular Roundhay Garden Scene, recorded with 12 frames per seconds and lasting just few seconds. This moment history was made and since then, technical improvements like resolution increase and ongoing price, weight and size decreases made camcorders mainstream and available for everyone in our modern society. Notable improvements of the camera technology are color support, the connection to Internet (IP cameras) and 3D support. Latest trend driven by camera technology is augmented reality, which in its first steps already existed in 1968. The success of modern cameras also depend heavily on software developments.. Statistical methods like perceptron algorithm and backpropagation accelerated the development of the field of computer vision (driven with the help of increasing computer power). Deep Learning became popular with improved learning algorithms starting in 2015 (Hinton). On the business model side, video cameras are the backbone of the million heavy movie and surveillance industry. In the social network area video sharing portals popped up making it possible for everyone to share video records. Figure 1 shows the brief overview of milestones in the area of video cameras in three different sections: Hardware, software and services.

II. Introduction

There is evidence[1] that human behavior improves, if humans are watched. This fact is causing endless political debates, on how much of privacy should be sacrificed to decrease crime rate. Video cameras play a leading role in that altercations, as they are installed in public spaces and an increasing number is placed in office buildings and housing complexes. Even more, smart video cameras are sold as consumer products and stream videos from housing spaces via cloud services onto the smart phones of their residents. With *Face ID*™and *Windows Hello*™, the cameras of consumer devices are always on. In sum, live streaming and live data processing video cameras are omnipresent. Some manufacturers already see added value in selling devices with build in webcam covers (figure 2) or allow to retract them (figure 3).

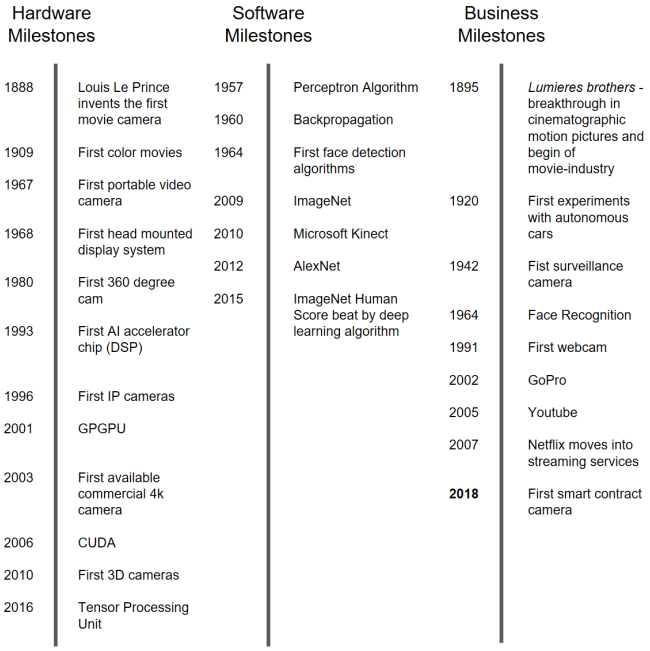


Fig. 1: Camera Milestones



Fig. 2: Webcam Cover

Source: Slashgear



Fig. 3: Recede mechanism

Source: Bosch

Taking into account actual developments in computer vision and block-chain allows to redesign the architecture and process on how smart cameras could work. Instead of going into the cloud to process visual information, smart cameras can analyze their data locally. And instead of struggling with proprietary solutions, we want to present an open, decentralized architecture for smart cameras. We propose a concept for video cameras to

- fully support **privacy** (III-A)
- implement smart contracts for **rights management** of camera and plug-in ownership (III-B)
- utilize a modular **plug-in approach** via an open protocol allowing **plug-ins to communicate** (III-C)
- implement **trigger**[1] **services** (III-D)
- integrate **marketplace mechanisms with token monetization** (III-E)
- to support wealth and future of society

(IV)

In chapter V we will list exemplary use cases realized with our open, decentralized and secure approach for utilizing secure smart cameras.

III. Block-chain enabled smart camera

A. Privacy mechanisms

We strongly believe that software based privacy protection is not enough and that physical based protection is crucial to maintain trust and increase willingness of users to use a camera, which is monitoring their activity in private or public environment. Spectre and Meltdown but also other popular bugs or hacks have shown the exceptional urgency to protect camera data. Our approach to improve trust level is twofold;

[1]The outcome of image/video detection models is always a probability output between [0-1]. This output is triggering an action. How actions are triggered, depend on underlying rules.

1) Visible physical protection with remained (locked) functionality: Using blocking mechanisms as shown in figures 2, 3 come with several drawbacks: (1) the camera is useless (2), users don't know if other sensors (like microphone) are still activated and (3) there is no standardized way to reactivate the camera remotely. Our design allows cameras to run locally and trigger signals on a *separate* data stream. This separate connection is build that way to make it very hard to transmit real image or video data.

2) Transparent and decentralized communication of dataflow: The communication of cameras (Plugins) to external enviornment (other users, other cameras or interfaces) is decentralized and managed via plugins.

1.1	Mainboard with storage for logics
1.2	Maincamera with slave trigger indicator
1.3	Touchdisplay representing User Interface
1.4	Indicator LED to represent privacy status
1.5	Measurepoint power supply
1.6	Indicators to represent logic trigger status

TABLE I: Mainsection

2.1	Fully transparent section
2.2	Attachable Databridge
2.3	Lockable access to install or uninstall databridge
2.4	Power supply via 2 visible copperbeams

TABLE II: Airgap section

3.1	Camera pointing on triggerindicators on mainunit
3.2	Logic Analyzer to receive triggers over airgap
3.3	Powerconnection
3.4	Ethernetconnection

TABLE III: Power section

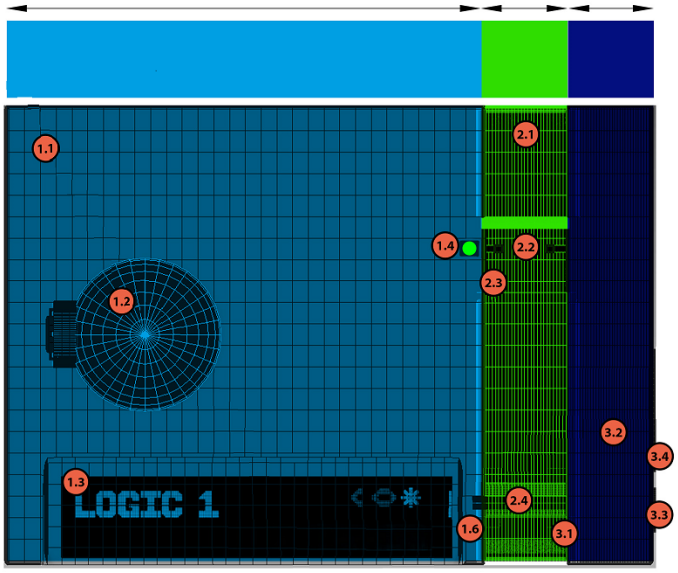


Fig. 4: Camera Concept part A

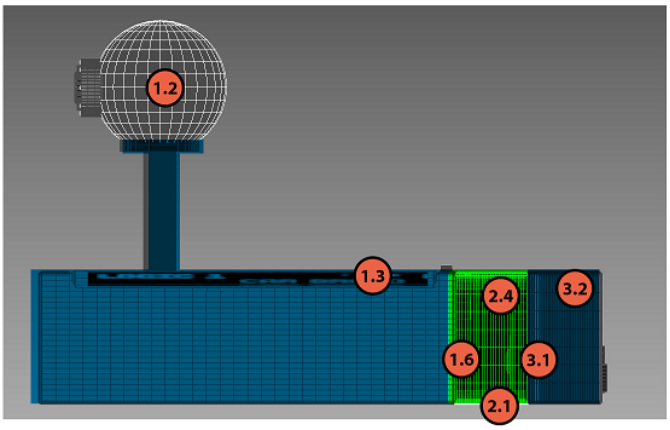


Fig. 5: Camera Concept part B

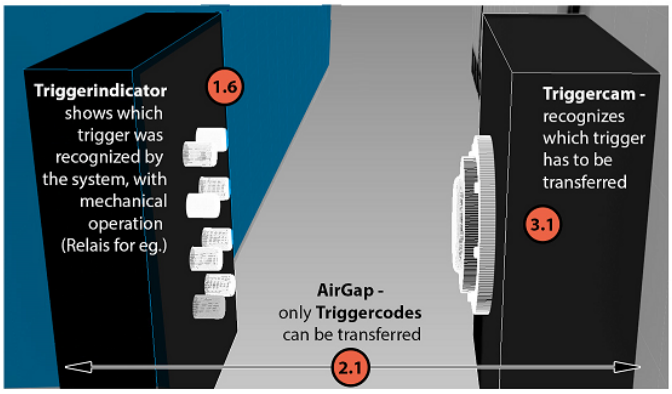


Fig. 6: Camera Concept part C

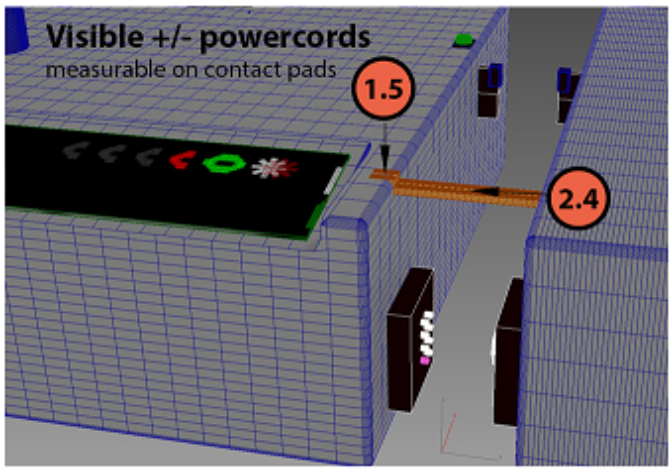


Fig. 7: Camera Concept part D

Figure 8 is listing used symbols in the following chapters. The camera symbol (C) is describing a video camera, the square (P) a plug-in for it, which can be an image detection model or a rule for triggers, the revenue model (R) shown as a square is depicting how the revenue is shared among users. The human icon (O) is representing persons or legal entities.

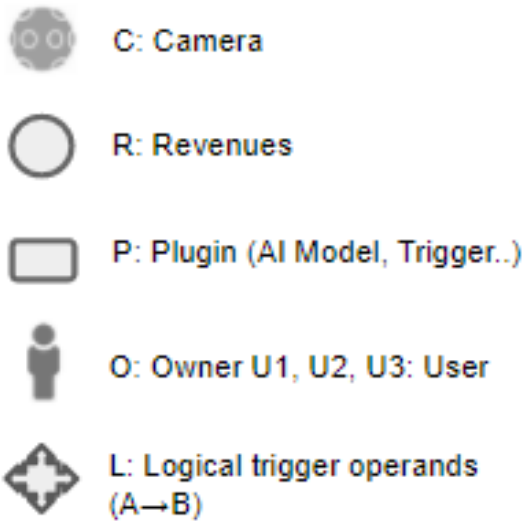


Fig. 8: Legend

B. Plugin Ownership and rights management

Figure 9 visualizes the plug-in and shared plug-in revenue architecture describing how plugins are managed.

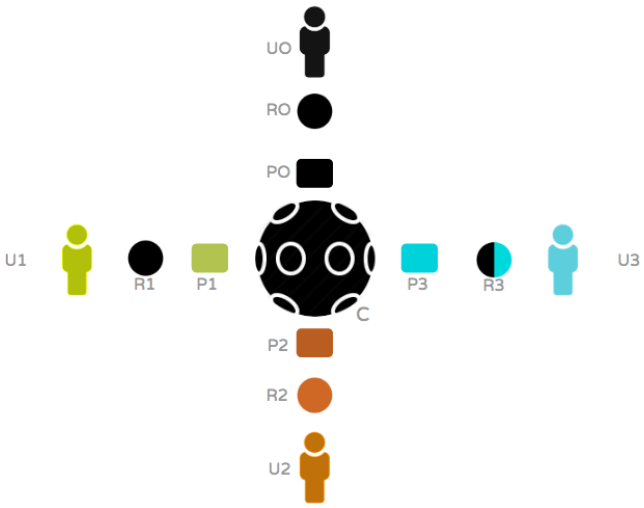


Fig. 9: Plugin Ownership

Four users are shown and a camera in the center. The black user (UO) is the owner of the camera, running a plugin or so named model (in terms of AI language) for his camera. This plug-in is enabling the camera to detect events, like if someone is in his home, or not. As he is the only owner of this plug-in, he is also the only revenue holder. So in case someone would like to know if someone is in his home, he can sell this information, all fees would go the owner. User 1 (U1) is gifting a plug-in to the owner (UO) and the owner remains the property owner. **Example:** Your insurance provider is gifting you a window and door open/closed model, helping you to automatically detect if your door gets opened (if you are not at home). As this information is valuable, but (most probably) not easy to sell, this plug-in can be seen just as a nice give a way. Another example is User 2 (U2). He is only willing to contribute a model if he can be the owner of the revenue stream. You agree, as this plug-in is detecting if your lights are up if you are not at home. Your energy provider is running a loyalty program and allows you to collect points, if you turn of your light if you are not at home. As you profit (indirectly) from this plug-in, you are willing to give ownership completely to your energy provider for this plug-in. User three (U3) shows the shared revenue stream of plug-in approach: User 3 wants to provide a model, but for agreeing it, the owner insists on a share to be negotiated. If both parties agree on the share, the plug-in can be installed on the camera. Example: A market research company wants to know, when and how long people in your flat are sitting in front of TV. As User 3 thinks this information is valuable and therefore can be sold to third parties, user 3 insists on a share of 50%. The market research company agrees on the share and the plug-in gets installed. Later on, a TV company also wants to directly access this information and sends a request to buy this information. The owner agrees and the incomes goes 50:50 to the owner and the plug-in contributor (research company)

C. Plug-in mechanism

1) Modular approach: 9 run several models from *different* frameworks in *parallel* to increase accuracy and make the quality of detection more robust.

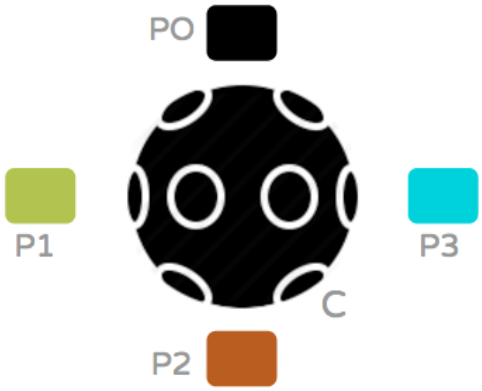


Fig. 10: Plugin Approach

2) Reusability: Focus of plug-ins is re-usability. Image detection but also trigger modules are build the way they run on almost every environment.

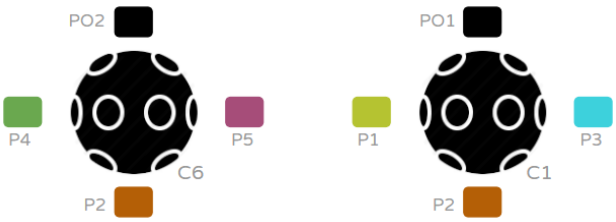


Fig. 11: Reusability Approach

D. Trigger

Due to increased capabilities of smart cameras to detect more and more visual events, a service to manage triggers (events) is crucial.

1) Triggers Logic: Figure 12 shows that external systems (users, interfaces) are only informed based on specific trigger logics.

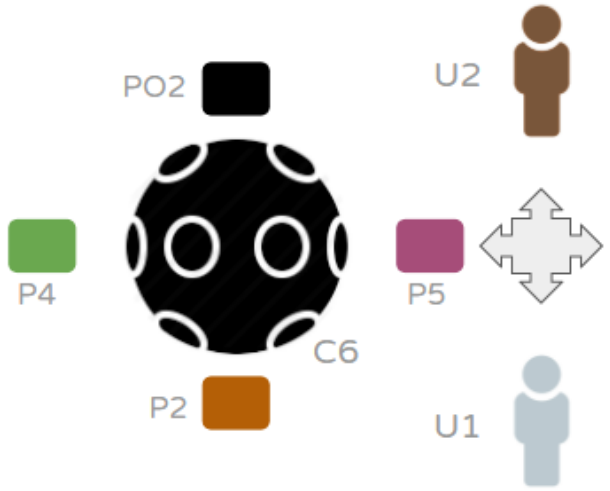


Fig. 12: Trigger Approach

2) Distributed Triggers: Figure 13 shows that two cameras are connected via distributed trigger-logic.

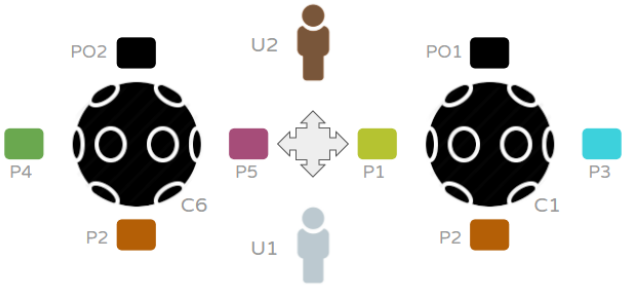


Fig. 13: Distributed Triggers Approach

E. The marketplace mechanism

We see data analysts and data owners (clients) in a complete diametrically opposed position. On the one side data analysts are challenged to deliver high accuracy, which is time consuming and expensive. Clients on the other side want a cheap and quick solution. A marketplace can challenge the creation of software but more importantly, is giving data scientists the chance to resell their software and build the AI software already in the beginning for a wider audience.

F. Block-chain mechanism

The blockchain mechanism describes 3 ways on how the block-chain is used in our concept.

1) Plugin Ownership - there will be an ecosystem of plug-ins
2) Camera Ownership - sharing camera ownership models help in commercialization
3) Token usage - (a) each camera can hold a public share which is paying out dividends on token holders (b) new cameras are bought with token revenues and are in 100% ownership of the community. External plugins can be contributed, the community is deciding in a democratic voting about plug-in policy.

G. Camera Ownership

Figure 14 shows that also the ownership of the camera can be changed and negotiated. The owner and user 2 both run their plug-ins on the camera of the owner. As the owner has a very good camera (=camera in a good context, he is running a restaurant and the camera is placed in that restaurant) user two strongly believes that many profitable plug-ins will run soon on that camera. For that reasons user 2 requests a share of that camera. The owner agrees on the negotiated fee and gives (in this example) a share of 50%. This means, all plug-ins which monetize have to be shared 50:50 of the negotiated part of a specific plug-in. In this example, user 3 gets 50%, the owner 25% and user 2 25%.

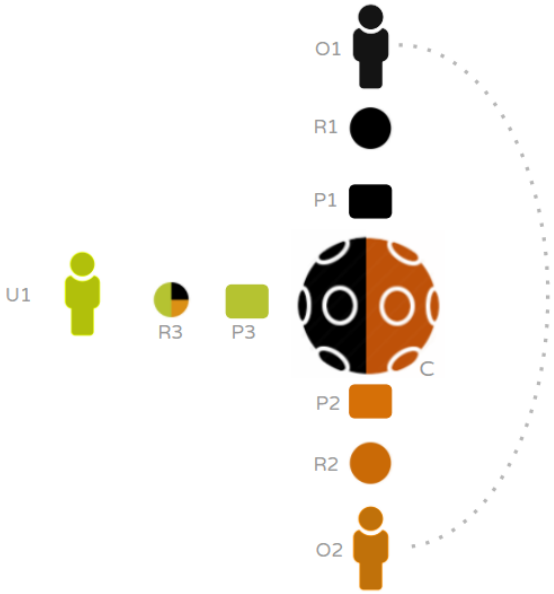


Fig. 14: Camera Ownership

IV. The Good Big Brother

Figure 15 shows the interest group principle. This means that teams have the possibility to focus on specific main areas like public press, medical aid, public surveillance, environmental issues and other fields to filter connected cameras and enable new services like media-drones which are objectively telling how many people are on a mass demonstration.

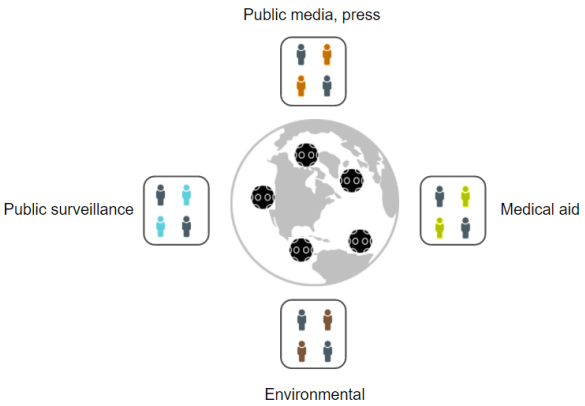


Fig. 15: Interest Groups

V. Use Cases

This architecture is enabling following use cases;

Use Case	Description	Example
Public interaction	Everybody can connect to the camera	A bypassing user wants to know whom the camera belongs to and what the camera is processing
Browsing Live Plug-ins	Browse smart cameras running specific plug-ins	Location based search of cameras which are running e.g. "Pet Models" to easily identify pet owners
Suggesting Plug-ins	Suggest and purchase remotely plug-ins to foreign cameras	Suggest a plug-in on a drone camera to count people on demonstrations and sell that information to media
Combining Plug-ins	Combine different plug-ins from different frameworks	Combine face detection from tensor-flow and face-detection from openCV
Request based information selling	Sell information collected by your camera on each request from a user	A user wants to get the information how many cars pass on average the street
Automated based information selling	Sell information collected by your camera automatically to users / interfaces	Several users wants to get the information how many cars pass on average the street you have placed the camera
Distributed Triggers	Define and run distributed triggers	Get informed if camera 2 sees 3 minutes later the same user on camera 3
Taking ownership shares	Get a share of a specific camera	Browse cameras which are highly economic and take a stake in ownership to receive dividends via token monetization
Taking ownership	Get full ownership of a specific camera	Buy the rights of a camera to install plug-ins at your will

References

[1] Nettle D, Nott K, Bateson M (2012) Cycle Thieves, We Are Watching You: Impact of a Simple Signage Intervention against Bicycle Theft. PLoS ONE 7(12): e51738. https://doi.org/10.1371/journal.pone.0051738